Machine Learning-Based Price Prediction System for Smallholder Farmers
Project Requirement / Specification

1. Project Overview

Agricultural price volatility is a major challenge faced by smallholder farmers. Farmers often lack access to reliable information about expected market prices, which leads to poor selling decisions and reduced income.

This project proposes the development of a machine learning-based web application that predicts crop prices using historical agricultural data. The system will utilize supervised machine learning algorithms to analyze past market prices and generate predictions that help farmers estimate future crop prices.

The platform will provide a simple interface where farmers can input crop-related information and receive predicted market prices, confidence levels, and price trends.

2. Problem Statement

Smallholder farmers frequently sell their crops without reliable information about future market prices. Due to this lack of predictive insight, farmers may sell their produce at unfavorable prices, resulting in financial losses.

Traditional methods of price estimation rely heavily on experience or market intermediaries, which may not always provide accurate information. Therefore, there is a need for a data-driven system capable of predicting crop prices using machine learning techniques.

3. Aim of the Project

To design and implement a machine learning-based price prediction system that helps smallholder farmers estimate future crop prices using historical agricultural market data.

4. Objectives of the Project

1. Collect and prepare agricultural price datasets from Kaggle.
2. Analyze historical crop price patterns using exploratory data analysis.
3. Implement machine learning models for crop price prediction.

4. Evaluate model performance using appropriate evaluation metrics.
5. Develop a web-based system that allows farmers to obtain predicted crop prices.
6. Compare different machine learning models and select the best-performing model.

5. Scope of the System

The scope of the system is limited to:
• Prediction of selected crop prices only
• Predictions based on historical market price data
• Use of public datasets obtained from Kaggle
• Predictions for selected markets
• Machine learning models trained offline
• A web-based interface for users to obtain predictions

The system does not perform real-time market data collection.

6. Target Users

Primary users of the system are smallholder farmers. Farmers will interact with the system through a simple web interface to obtain predicted prices for selected crops.

7. System Inputs

The system requires the following user inputs:
• Crop Type (e.g., maize, rice, tomato)
• Market Location
• Date / Month
• Previous Price
• Weather Information (optional)

Example input:
Crop: Maize
Market: Abuja
Month: August
Previous Price: ₦350

8. System Outputs

The system will generate:
• Predicted Price
• Confidence Level
• Price Trend

Example output:
Predicted maize price: ₦420

Confidence: 87%
Price trend: Increasing

9. Data Source

The system will utilize public agricultural datasets obtained from Kaggle. These datasets typically include fields such as:

- Crop
- Market
- Date
- Min Price
- Max Price
- Modal Price

The datasets will be cleaned and processed before machine learning training.

10. Machine Learning Models

The following supervised machine learning algorithms will be implemented:

- Linear Regression (Baseline Model)
- Decision Tree Regression
- Random Forest Regression

The models will be compared and the best-performing model will be selected for the final prediction system.

11. Model Evaluation Metrics

Models will be evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Error)
- R^2 Score

12. System Architecture

System components include:

1. User Interface – Web form for farmer inputs
2. Backend Server – Processes user requests
3. Machine Learning Model – Generates predictions

Workflow:
User Input → Web Application → Backend Server → Machine Learning Model → Predicted Price Output

13. Technology Stack

Programming Language: Python

Machine Learning Libraries:
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Backend Framework:
- Flask or FastAPI

Frontend:
- HTML
- CSS
- JavaScript

Development Environment:
- Jupyter Notebook / Python IDE

14. System Constraints

- Predictions depend on dataset quality.
- Only historical data is used.
- No real-time market updates.
- External economic factors may affect price accuracy.

15. Expected Contributions

- Demonstrate the use of machine learning for agricultural price prediction.
- Provide a simple decision-support tool for farmers.
- Show how predictive analytics can support agricultural markets.